Assessing key traits to promote overyielding in mixtures of legumes and non-legumes: A case study using the Virtual Grassland model

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***Abstract*— The role of various plant traits involved in carbon (C) and nitrogen (N) economy on species balance and total aboveground biomass production in binary mixtures of legumes and non-legumes was evaluated through a modelling approach using the Virtual Grassland model (VGL). A first step allowed us to identify the model parameters most sensible to inter-specific competition trough a sensitivity analysis. A second step consisted in testing the impact of individual traits values (i.e. selected among sensible parameters) on virtual mixture performance. Based on our simulation results, we concluded that maximal overyielding was achieved in cases where trait values were divergent for N acquisition (i.e. allowed complementarity in the use of different N pools) but convergent for light interception (i.e. limiting the asymmetric competition for light). The best combination of traits was not the same in all the pedo-climatic conditions tested and depended on the level of mineral N available into the soil. Random trait combinations could frequently lead to reduced mixture yields (as expected from a neutral situation of competition) and even to under-yielding (i.e. less than the average of monocultures) in some situations.**

Keywords—Individual-based model, legumes, mixtures, overyielding, trait divergence, complementarity, competition

I. Introduction

In order to deal with the challenges of agroecology, there is a considerable need for quantitative models able to link plant traits with ecosystem functioning and plant performance in inter-cropping systems [1, 2]. This is especially true in multi-species grasslands typically promoted in Europe for a sustainable production of forages for ruminants [3]. For this purpose, individual-based models (IBM) of plant communities appear as promising tools to improve our understanding of inter-plant competition, both above- and belowground [4, 5].

The 'Virtual Grassland' initiative (VGL) was launched in this context [6]. It proposes an individual-based model of grassland communities that integrates the plastic responses of plants to light, water and nitrogen. Perennial grass and legume species are the two functional groups which usually dominate sown grasslands, and are each considered using specific plant models [7, 8]. These two models describe plant morphogenesis at the phytomer level as resulting from the functioning of plant meristems in response to external and endogenous environments. By integrating the local responses of plants above- and below-ground, VGL aims to scale plant-plant interactions at the population and community levels in order to simulate changes in grassland composition over time in response to climate and management [9].

Yield benefits from legume-based forage mixtures have long been acknowledged and were related to improved resource use [10, 11]. Consequently, the total yield of such mixtures quite often exceeds that of the average of monocultures (i.e. overyielding), and can also exceeds that of the best monoculture (i.e. transgressive overyielding) in low-inputs environments [12]. The role of niche separation between species for nitrogen (N) acquisition and the ability of legumes to fix and transfer atmospheric N are clearly favoring these effects [13, 14]. The greater spatial and temporal complementarity offered by mixing species to capture light and other resources was also suggested [10]. However, little is known about the component traits that might have the most significant influence. As N fixation and photosynthetic capacity are tighly related within plant communities, any trait involved in the competition for light and N could contribute [15, 16, 17]. This lack of quantitative references still hampers our ability to choose the most suitable plant phenotypes to associate among existing species and cultivars [18]. The rationale behind current assembly rules are indeed either purely empirical [19] or driven by general principles seeking to maximize trait divergence between species [20]. Plant community IBMs could help to decipher the role of the different traits in the mixture's performance and make the whole process of assembling species and cultivars more efficient.

In this study, we evaluated through a modelling approach the role of various plant traits involved in C and N economy on total aboveground biomass production by binary mixtures of legumes and non-legumes. To do so, the VGL model was used under three scenarios of N management. The consequences of maximizing trait divergence for C and N acquisition and use were analyzed in these contrasting situations.

II. Materials and Methods

A. Overview of the model

The VGL plant model dedicated to legume population dynamics was used in this study ("L-egume" model, [8]). It is based on the L-system formalism [21] and deals with 3D shoot and root morphogenesis for contrasting morphotypes of dicotyledoneous plants, as well as C, water and N exchanges with the environment. To do so, the plant model is coupled within VGL with two environmental sub-models dealing with radiation transfer and light competition aboveground [22] and soil water and mineral N balances belowground [23].

The L-system uses a daily time step and computes the potential morphogenesis of vegetative shoots and roots as a result of the functioning of plant meristems and growing tissues. It encompasses two feedback loops to account for plant responses to their local environment and to regulate actual growth and morphogenesis by light [24] and N availability [25]. Total plant dry matter production is determined from light interception using a radiation-use-efficiency approach [26]. Biomass allocation between shoots and roots is assumed to follow an allometric relationship, whereas biomass partitioning between shoot organs (i.e. leaves, stems) is driven by the relative carbon demands of individual organs. A detailed description of the model equations can be found in Louarn and Faverjon [8] for the plant model and Brisson et al. [27] for the soil model. A total of 43 equations with about 120 parameters are defined in the plant model. Only the principal parameters involved in plant morphogenesis and C and N use are summarized in Table 1.

B. Sensitivity analysis

A first step in determining the major traits involved in inter-plant competition was to perform a sensitivity analysis of the plant model in binary species mixtures. To do so, a reference plant morphotype was defined as a target competitor. The G-morphotype (i.e. erected crown-former similar to alfalfa) described in [8] was chosen, as it appeared as a good candidate to represent productive grassland species under low N environments that are relevant to legume-based grasslands. A simple approach involving the change of one-parameter-at-a-time (OAT) was used to define a range of virtual companion species and quantify the effects of parameter values in the companion species on model outputs [28]. An important step thus consisted in determining the nominal values and ranges of model parameters. The nominal values were taken equal to the G- morphotype, thus defining a neutral situation of competition where competitors were identical and grew proportionally to their relative density (Fig. 1A). Then, for 28 parameters involved in C-N acquisition and use, we synthesized several publications ([29, 30] for morphogenetic parameters; [25, 27] for C and N use) in order to find an adequate range of variation for each parameter (Table 1). A total of 5 step values were regularly sampled within this range of variation to run simulations. Sensitivity of model output was then measured by monitoring changes in the output by linear regression. During this phase of screening, no interaction between parameters was considered and only the main effect of each parameter on the variance of model outputs was computed. Two main outputs were targeted in this phase: the total annual production of the community and the relative yield of the companion species in a mixture sown at 50/50%. For the purpose of comparison, the same analysis was performed on a pure stand of the reference target morphotype (100% G- morphotype).

During this exercise, the simulations were performed for dense plant stands (400 plants.m^{-2}) in the first year of production after stand establishment. The plot location was typical of Western European temperate grasslands (Lusignan, 46.26°N, 0.11°E) and we considered 30-year average weather data (1986-2016) as meteorological inputs for the model. The edaphic conditions used as inputs corresponded to a 1.5-m-deep soil with 1.1 g organic N.kg^{-1} in the upper soil layer. Under low to moderate N fertilization, this soil usually produces N-limiting growing conditions for non-fixing grassland species, even when grown in mixture [31]. Here, the applied management allowed a moderate N fertilization of 30 kg N.ha^{-1} during each regrowth (i.e. 120 kg N.ha^{-1}.y^{-1}) as well as an irrigation schedule matching potential evapotranspiration and crop water requirements. A total of four harvests were performed during the year according to a fixed schedule (at days of the year number 189, 230, 283 and 335).

C. Virtual experiments

A second step in analyzing the role of major plant traits in the performance of grassland mixtures was to build a virtual experiment testing binary mixtures varying in their trait value composition and systematically assembling a legume morphotype (i.e. FIX^{max} always set to 24 g N.kg^{-1}) with a non-fixing (i.e. FIX^{max} always set to 0 g N.kg^{-1}) morphotype (Fig. 1C).

In order to test the relevance of assembly rules based on a maximal trait divergence, the role of major plant traits controlling four physiological functions (i.e. light acquisition, mineral N acquisition, kinetics of growth and resource use efficiency) involved in the niche separation of species were examined. Based on the results of the sensitivity analysis, four model parameters associated with these four physiological functions were selected among the 28 parameters previously tested. They corresponded to the parameters displaying the highest effect on model outputs in mixture for each physiological function. The virtual experiment then consisted in testing all the possible binary mixtures involving the combination of these four traits in a complete factorial design with three levels allowed for each trait (highest differential in favor of the legume, equal, highest differential in favor of the non-legume). This defined a total of 81 possible mixtures, with one of them corresponding to a null model of competition where morphotypes differed only by their ability to fix atmospheric N (Fig. 1C). As the outcome of competition experiments is usually affected by intra- and inter-specific competition, and thus sensitive to both absolute and relative plant densities [32], each binary mixture was analyzed though a whole set of simulations at high density considering different species proportions along the classical de Wit's replacement design [33]. Seven relative densities were tested for each species (i.e. 100%, 87.5%, 75%, 50%, 25%, 12.5% and 0%). This allowed us to properly disentangle the effects of plant traits, initial species proportions and actual species proportion

on the overall annual production of the community. In particular, it was possible to compute different indices characterizing resource acquisition and mixture production, such as the average mixture over-yielding over the whole possible range of sowing proportions, or the partial yields of each species at a 50/50% sowing.

During this virtual experiment, the simulations were performed using the same location, climate, soil and management as in the sensitivity analysis. However, in order to test the dependency of the conclusions to soil N fertility, three different scenarios of mineral N application were tested that corresponded to zero (N0), moderate (N120, 120 kg $N.ha^{-1}.y^{-1}$) and high (N400, 400 kg $N.ha^{-1}.y^{-1}$) fertilization rates.

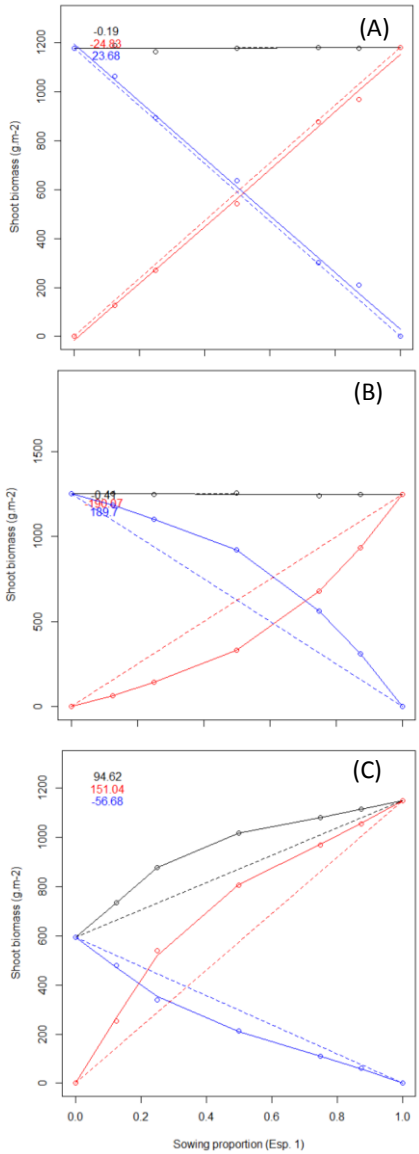

Fig. 1. Variation of total yield (black) and partial yields (red and blue) simulated for virtual binary mixture tested in the Wit's replacement design which makes species proportion vary from 0% to 100%. The first panel (A) corresponds to a neutral situation of competition where the two species present exactly the same trait value, B) the two species differ by a single trait value without effect of total yield at any proportion (i.e. L^{max}_{In}) and C) the two species differ by a single trait affecting N fixation (FIX^{max}). The dashed lines represent the yield that would be achieved proportionally to the pure crop yields. The colored texts indicate the average yield differential achieved with respect to this theorical line (i.e. overyielding).

TABLE I. LIST OF PLANT PARAMETERS USED IN THE SENSITIVITY ANALYSIS AND THEIR RANGE OF VARIATION. LIGHT GREY, GREY, DARK GREY AND WHITE BACKGROUNDS INDICATE PARAMETERS ASSOCIATED WITH SHOOT MORPHOGENESIS, ROOT MORPHOGNENESIS, CN USE, AND N STRESS RESPONSES, RESPECTIVELY.

Symbol	Plant parameters		
	Description	***Unit***	***Range***
$Del50_{In}$	Delay for the onset of internode growth within a phytomer	phyllochron	0 / 3
$Del50_L$	Delay for the onset of leaf growth within a phytomer	phyllochron	0 / 1
Elv_L	Average elevation angle of the leaves	°	20 / 70
L^{max}_{In}	Maximum internode length	cm	2.5 / 6
L^{max}_L	Maximum leaf length	cm	2 / 5.5
nsh^{max}	Ceiling number of primary axes	-	4 / 50
$Phyllo_1$	Phyllochron for the primary axes	°Cd	20 / 60
q	Parameter controlling optimal temperature	-	1.5 / 3.2
$SPAN_L$	Leaf life span for isolated plants	Phyllochron	10 / 30
Tilr	Tillering rate	$phyllochron^{-1}$	0.2 / 1.2
$Alloc_{FR}$	Partitioning coefficient between fine roots and taproot	$g.g^{-1}$	0.10 / 0.75
D^{min}	Minimum root apex diameter	cm	0.01 / 0.03
D^{max}	Maximum root apex diameter	cm	0.04 / 0.25
Dldm	Average ratio between lateral and parent root diameters	-	0.1 / 0.45
$Elmax_R$	Elongation rate of roots at the maximal apex diameter	$cm.°Cd^{-1}$	0.07 / 0.25
IBD	Average inter-branch distance	cm	0.20 / 0.40
GDs	Root growth duration parameter	$°Cd.mm^{-2}.mg^{-1}$	1000 / 2500
LDs	Root life span parameter	$°Cd.mm^{-2}.mg^{-1}$	30000 / 100000
RTD	Root tissue density	$g.m^{-3}$	0.03 / 0.12
FIX^{max}	Maximum plant fixation rate	$g\ N.kg^{-1}$	0 / 36
NODcost	Relative reduction of the RUE for a 100% fixing plant	-	0.05 / 0.35
RUE	Whole plant radiation use efficiency by non fixing plants	$g.MJ\ PAR^{-1}$	1.8 / 3
$VMAX_1$	Maximum rates of absorption achieved by the HATS	$\mu mole.cm^{-1}.°Cj^{-1}$	0.0018 / 0.018
$VMAX_2$	Maximum rates of absorption achieved by the LATS	$\mu mole.cm^{-1}.°Cj^{-1}$	0.05 / 0.5
$KMAX_2$	Michaelis affinity constant for the LATS	$\mu mole.cm^{-1}$	25000 / 100000
$NNI50_{growth}$	NNI at which a 50% reduction of organ growth occurs	-	0.4 / 0.6
$NNI50_{dev}$	NNI at which a 50% reduction of axis development occurs	-	0.4 / 0.6
$NNI50_{photo}$	NNI at which a 50% reduction of photosynt occurs	-	0.3 / 0.5
$ZETA^{tr}_{esh}$	ζ threshold inducing a cessation of axis development	-	0.25 / 0.45

III. Results

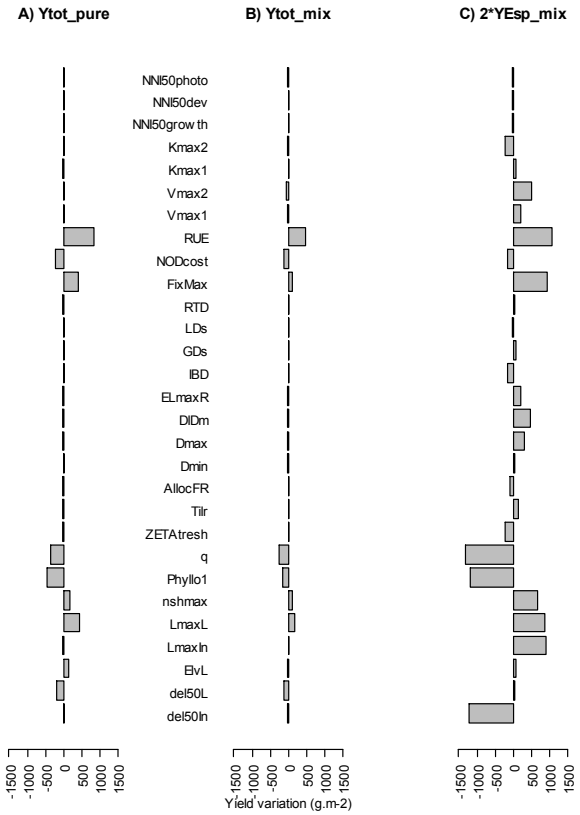

Fig. 2. Main effect of individual plant parameter on the variation of total annual yield in monoculture (A), total annual yield in binary mixture (B) and partial yield of the tested companion species (C)

A. *Parameter sensibility in pure stands*

The parameters most affecting the total annual yield in pure stands (Fig. 2A) were related to light acquisition process and the kinetics of development of the canopy aboveground (q, phyllo1, LmaxL), as well as with the efficiency of resource use (RUE, FixMax). Despite a moderate availability of soil N, a greater overall sensitivity of crop productivity to parameters involved in the morphogenesis of shoots could be noticed as compared to root morphogenesis, root functioning and N stress response parameters.

B. *Parameter sensibility in binary mixtures*

In binary mixtures, the total annual yield of the community (Fig. 2B) showed a pattern of sensitive parameters strikingly similar to that of pure stands. The same four parameters displayed the strongest influence on total yield (i.e. q, phyllo1, LmaxL and RUE). For all the parameters however, the magnitude of the yield variation appeared lower in mixture than in pure stands. This may indicate an ability of the reference species to buffer any change in the productivity of the tested companion species.

By contrast, the partial yields of the companion species in mixture (Fig. 2C) showed a much larger range of variation than total annual yield to the imposed changes in model inputs. Interestingly, not only the magnitude, but also the relative sensitivity of parameters changed when analyzing the balance between species. Indeed, some parameters, which had low or no impact on total annual yield, could deeply affect the balance. This was the case for instance regarding parameters involved in stem elongation (LmaxIn, Del50In), root elongation (DlDm, Dmax) and N acquisition (Vmax2, MaxFix). Overall, parameters involved in the morphogenesis of shoots and in the efficiency of resource use still had the strongest impact, but some of those accounting for root morphogenesis and root functioning could affect the species balance significantly, too. The parameters involved in N stress responses had little effect on model outputs under the conditions tested.

C. *Impact of soil N fertility on the neutral situation of competition between legumes and non-legumes*

As a result of the previous sensitivity analysis, four model parameters were selected to affect plant trait variation along phenotypic axes representing the physiological functions of light acquisition (LmaxIn), mineral N acquisition (Vmax2), kinetics of growth (q) and resource use efficiency (RUE) in binary mixtures. Figure 3 presents the situation of perfect trait convergence in a binary mixture between a legume and a non-legume species (i.e. all parameter values equal except for FixMax) grown at different rates of N fertilization. At the two extreme sowing proportions (100/0; 0/100), it was apparent that the productivity of the legume crop in pure stands was higher than the productivity of the non-legume crop in pure stands, irrespective of N availability. The annual productivity of the pure legume species increased 1.3-fold between N0 and N400, whereas it increased 2.7-fold for the non-legume from about 600 g.m-2 to 1600 g.m-2. The difference of production between the pure species was very low at N400 (about 50 g.m-2), suggesting that periods of significant N-stress almost never occurred in this treatment.

Quite remarkably, this neutral mixture systematically achieved an overall dry matter production that was higher than expected from the weighted average of the monocultures. The average overyielding was maximum in absence of fertilization (106 $g.m^{-2}$ at N0), still significant under moderate N (60 $g.m^{-2}$ at N120), but very low under high N (18 $g.m^{-2}$ at N400). In all the situations, overyielding resulted from an increase of the legume production which overcompensated the simulated reduction of production by the non-legume. Thus overyielding was also associated with an imbalance in the mixture that tended to switch towards the domination of the legume species under low N. A complementary analysis of the mixture production with respect to the species proportions actually realized indicated no significant overyielding (not shown). This suggests that sampling of the most suited sown species, rather than resource use complementarity, may explain overyielding in this situation.

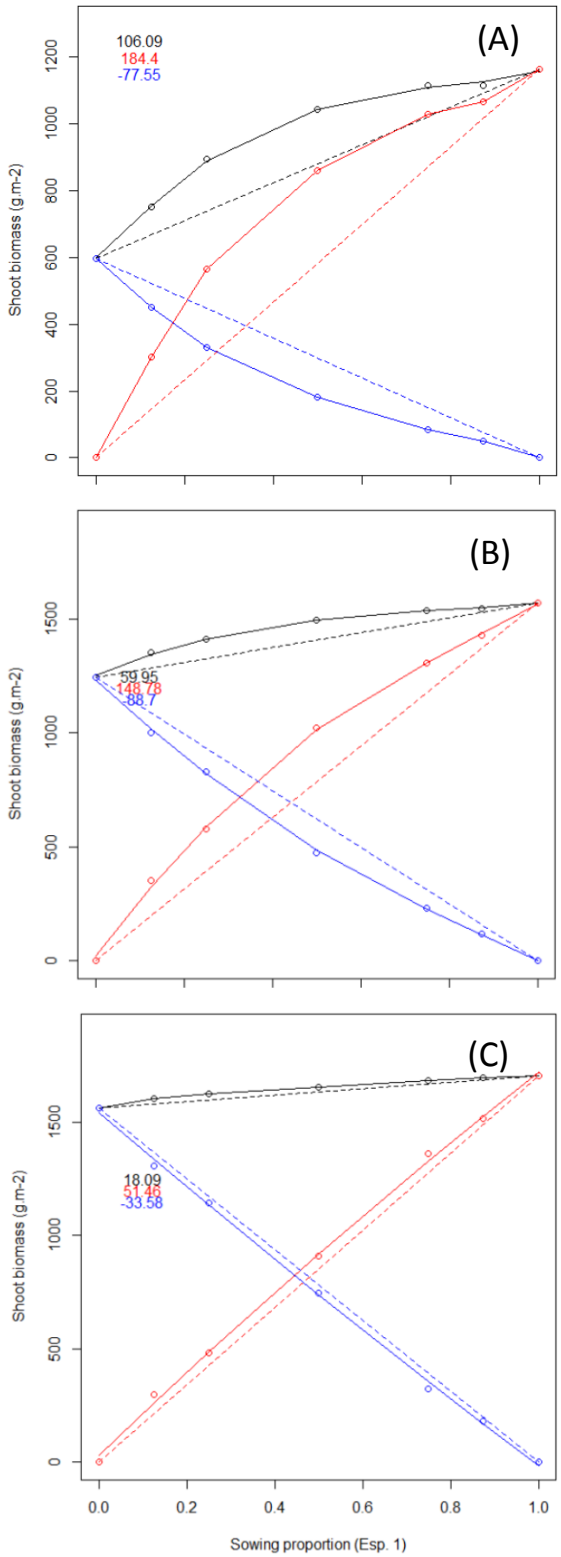

Fig. 3. Variation of total yield (black) and partial yields of species for a binary mixture representing a neutral situation of competition between a legume (red) and a non-legume (blue) (i.e. exactly the same trait value, except for N fixation) tested in the Wit's replacement design. Panels A), B) and C) are for N0, N120 and N400 treatments, respectively. The dashed lines represent the yield that would be achieved proportionally to the pure crop yields.

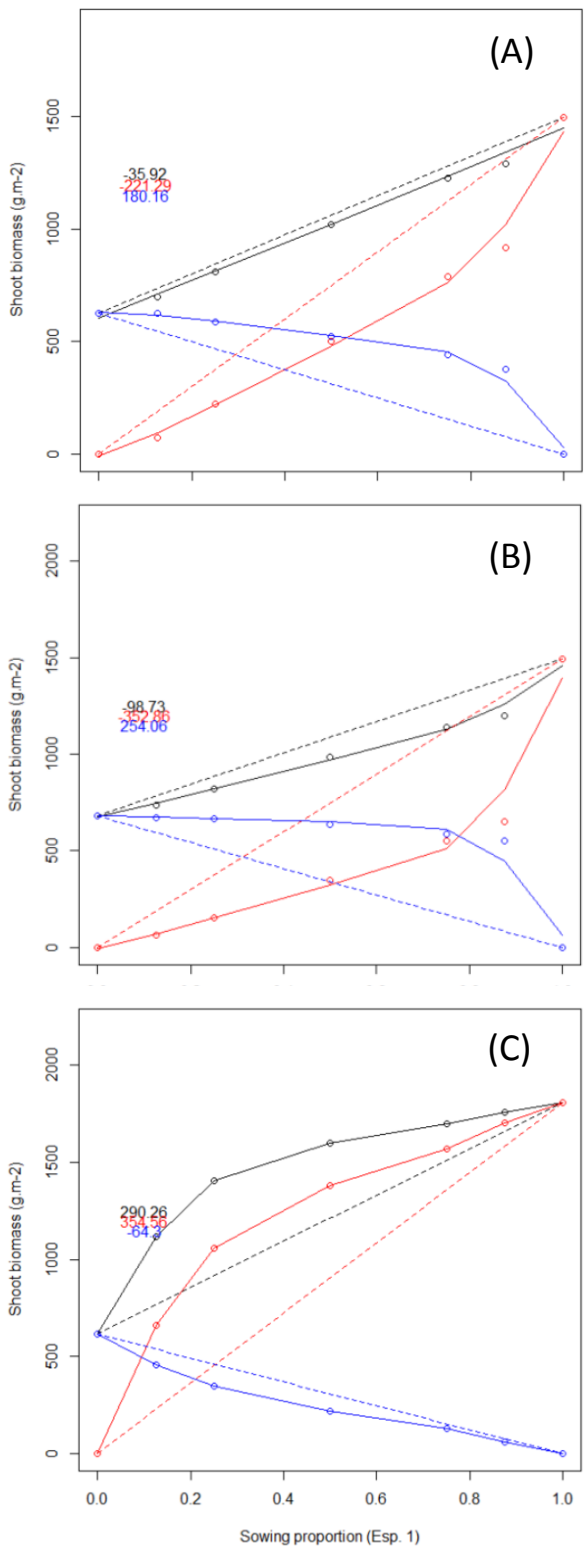

Fig. 4. Variation of total yield (black) and partial yields of legumes (red) and non-legumes (blue) for three binary mixtures tested in the Wit's replacement design at low N availability (N0). Panel A) represents a synchroneous mixture with trait diverging for mineral N acquisition and CN use. Panels B) and C) represent the corresponding asynchroneous mixtures with early non-legume and early legume, respectively. The dashed lines represent the yield that would be achieved proportionally to the pure crop yields.

D. *Impact of trait divergence on the balance between species and on the overall productivity of binary mixtures between legumes and non-legumes*

Departures from the neutral situation of competition induced by plant trait variations were analyzed for each growing conditions. Figure 4 illustrates the results for a series of mixtures differing by their traits for light and N acquisition. As a general picture, trait divergence favoring the acquisition of different resources by different species resulted in more balanced mixtures, whereas the opposite (i.e. trait value divergence always favorable to the same species) quickly lead to unbalanced mixtures dominated by a single species.

Thus, given the intrinsic advantage of legumes at low N, the use of non-legumes dominant for light or mineral N acquisition contributed to establish more balanced plant communities. In most situations, however, this evenness was achieved at the expense of a reduction of the overall production of the mixture (e.g. Fig. 4AB). This was apparent with overyielding scores that were lower than the neutral situation of competition (Fig. 3B). For instance, an improved competitive ability of the non-legume for light interception had a more negative impact on the growth of legumes than it had a positive impact on the growth of non-legumes (not shown). Regarding traits controlling mineral N uptake, the advantage diverging trait values depended on the level of N availability. At N0, the possibility of niche differentiation was limited by the amount of N into the soil. Under these conditions, non-legumes were not able to valorize their relative uptake advantage in terms of their own growth. This could even result in mixtures that produced less than expected from the weighted average of the monocultures (Figure 4B). At intermediate N levels (N120), complementary resource use could be observed and it was valorized through slight increases of overyielding (up to 78 $g.m^{-2}$, as compared to 60 $g.m^{-2}$ for the neutral situation) with an improved balance between species. At N400, N traits had little influence on the overall production (i.e. as previously noticed for the neutral model, Fig 3C).

The kinetics of growth affected both light and N acquisition. The temporal variation of growth had the strongest impact on the potential production of pure crops, as well as on the balance and overyielding of mixtures. An early seasonal growth usually resulted in a higher annual productivity, irrespective of N treatments. Asynchronous mixtures hosing such species displayed both the best and worst agronomic performances. They amplified the variations previously presented for light and mineral N competition in the corresponding synchronous mixtures. The temporal complementarity between species thus appeared as generally favorable to overyielding when using early legumes and detrimental to overyielding with the use of early non-legumes (Figure 4BC). In the simulated situations, the maximal overyielding within an acceptable range of balance (i.e. 25/75) was achieved at N120 for a mixture with convergent traits for light acquisition, and divergent traits favoring non-legumes in mineral N acquisition and resource use efficiency and favoring legumes in terms of growth kinetics.

IV. Discussion

Despite obvious limitations of such a modelling exercise [34, 35], the heuristic value of IBMs to infer plant-plant interactions can assist us in reinforcing logical and quantitative thinking in the design of assembly rules for intercropping systems [4, 36, 37]. For instance, the assumption of maximal trait divergence within a local pool of species is often promoted as a rule of thumb to maximize the complementarity between species in sown grasslands [20]. Several general conclusions can be drawn from our work on binary mixtures that only partially support this assumption:

- First, plant community models allowed us to define a neutral situation of competition against which it was possible to confront the effects of any trait combination or management practice [38]. Regarding binary mixtures of legumes and non-legumes, it allowed us to confirm that overyielding is an intrinsic property of such mixtures, even in the absence of trait divergence (other than N fixation). A sampling effect, rather than resource use complementarity, was the cause of overyielding in the neutral situation [39].
- Second, the impact of trait divergence could enhance overyielding at the same time as favoring a good balance between species in situations where it effectively resulted in a niche separation of the species for different resources. This was the case for mineral N acquisition in favor of the non-legume at N120. This was also the case for temporal complementarity of growth in favor of the legume.
- However, trait convergence, rather than divergence, was the most favorable situation regarding light competition. This was consistent with several reports indicating that dispersion in shoot growth traits could induce competitive exclusion and reduce diversity in experimental communities [40, 41, 42].
- Furthermore, it was apparent from the couple of complementary traits tested that the effect of divergence was generally not symmetric when applied to one or the other species. This was previously shown for light interception in pea-wheat mixtures [16]. Here we confirmed that given the functional role and higher potential of legumes in mixtures at low N, trait variations favorable to this component generally had a higher positive impact on the community productivity (e.g. for temporal growth). This implies that a naïve application of trait divergence that would ignore the functional and agronomic peculiarities of species is likely to hinder the advantage of trait diversity in cultivated mixtures.
- Finally, our results confirmed that the best combination of traits is likely to change with the environment and management of plant stands [42]. For instance, depending on the level of mineral N available, trait divergence for mineral N acquisition could lead to complementarity (N120), increased competition (N0) or neither competition nor complementarity between species (N400).

The present results were obtained for the first year of production after stand establishment. During this period, facilitation effects usually remain low because net N transfers from the living legume to the non-legume are barely significant [43]. As a consequence, complementarity and sampling effects explained most of the positive effect of legumes on mixture overyielding. This is well representative of annual crops intercropping and perennial grassland mixtures in their first year [31, 44], but does not give a complete and comprehensive overview of grassland functioning in the following years [31]. Thus, the present work should be completed with a similar analysis of the mixture in the following years, when significant N transfers do occur.

Plant community IBMs are still unlikely to propose accurate predictions of community dynamics on the long term for many pedo-climates any time soon. Their calibration for a particular set of plant diversity and their initialization for a given soil (e.g. spatialized soil variables) remain important issues to solve. Nevertheless, this study performed with the VGL model demonstrated that they could already offer a useful quantitative framework to test hypotheses and make 'what if' projections. For instance, VGL could help us to sort the most sensitive traits and plant functions in a particular mixture and assist us in choosing the sense and range of variation susceptible to produce favorable effects.

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